

● EASY

● ALGEBRA

Linear functions

02

10 questions · ~15 min

Q1

ALGEBRA

For the linear function f , the graph of $y = fx$ in the xy -plane has a slope of $\frac{1}{4}$ and passes through the point $(0, 5)$. Which equation defines f ?

A. $fx = \frac{1}{4}x + 5$

B. $fx = \frac{1}{4}x + \frac{1}{5}$

C. $f(x) = \frac{1}{4}x - \frac{5}{4}$

D. $fx = \frac{1}{4}x - 5$

A contract for a certain service requires a onetime activation cost of \$ 35 and a monthly cost of \$ 23. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

A. $c = \frac{t}{23} + 35$

B. $c = \frac{t}{35} + 23$

C. $c = 23t + 35$

D. $c = 35t + 23$

Robert rented a truck to transport materials he purchased from a hardware store. He was charged an initial fee of \$20.00 plus an additional \$0.70 per mile driven. If the truck was driven 38 miles, what was the total amount Robert was charged?

- A. \$46.60
- B. \$52.90
- C. \$66.90
- D. \$86.50

$$T = 1,000 + 18h$$

In the equation above, T represents Brittany's total take-home pay, in dollars, for her first week of work, where h represents the number of hours she worked that week and 1,000 represents a sign-on bonus. If Brittany's total take-home pay was \$1,576, for how many hours was Brittany paid for her first week of work?

- A. 16
- B. 32
- C. 55
- D. 88

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $fx = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of 33 in this context?

- A. The estimated height, in cm, of the liquid at the start of the project
- B. The estimated height, in cm, of the liquid at the end of the project
- C. The estimated change in the height, in cm, of the liquid each day
- D. The estimated number of days for all of the liquid to evaporate

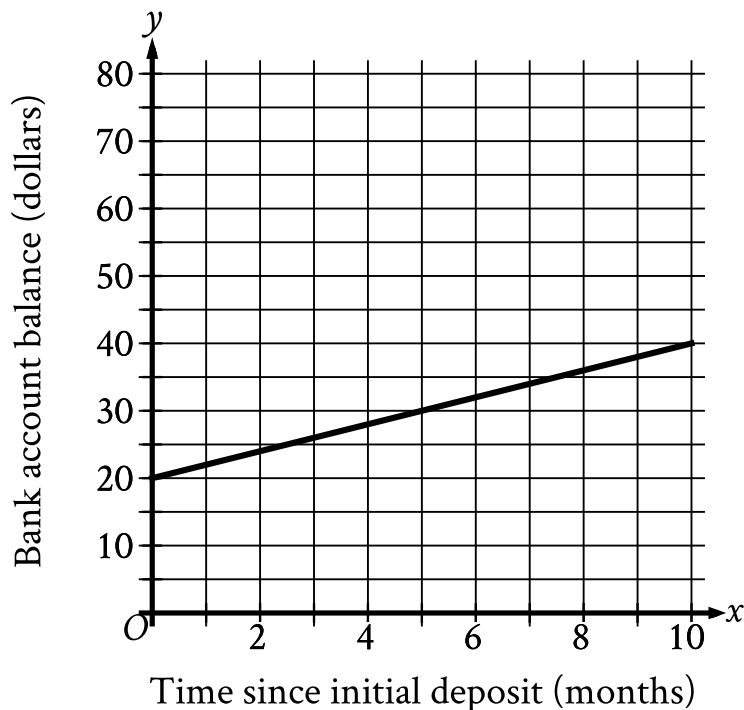
The function g is defined by $gx = 4x - 6$. What is the value of $g-7$?

A. -34

B. -22

C. $-\frac{13}{4}$

D. $-\frac{1}{4}$



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (0 comma 20)
 - (5 comma 30)
 - (10 comma 40)

A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

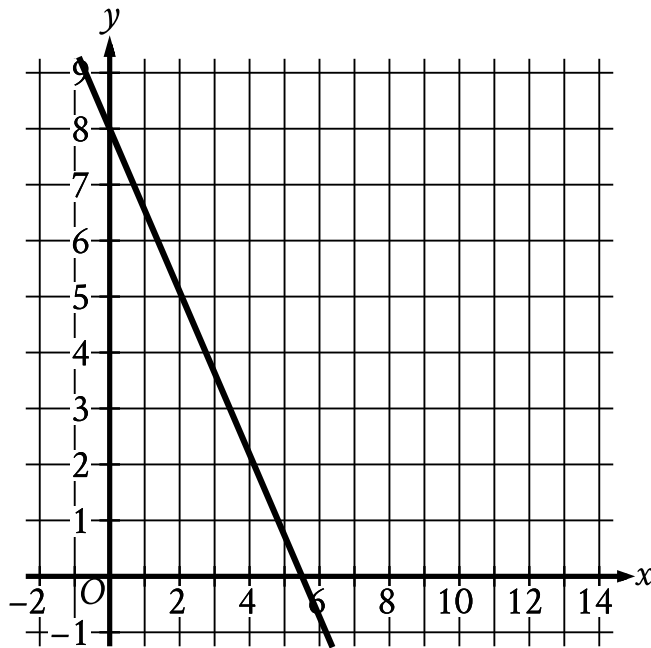
The graph of the function f is a line in the xy -plane. If the line has slope $\frac{3}{4}$ and $f(0) = 3$, which of the following defines f ?

A. $f(x) = \frac{3}{4}x - 3$

B. $f(x) = \frac{3}{4}x + 3$

C. $f(x) = 4x - 3$

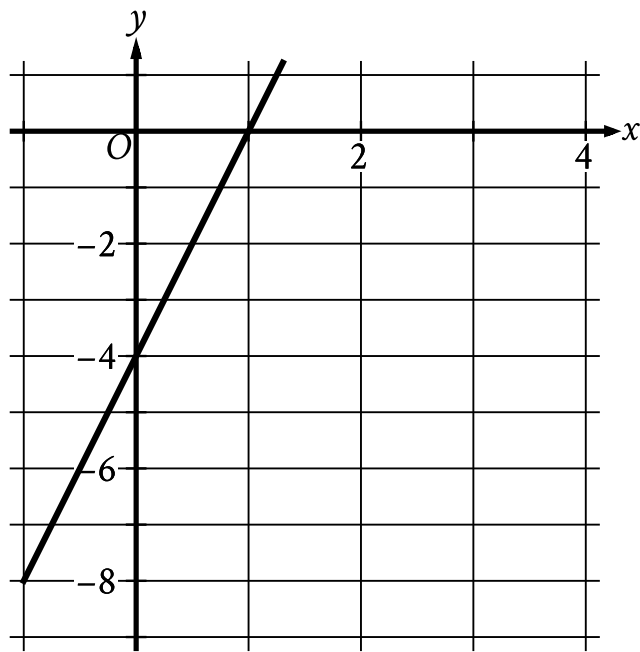
D. $f(x) = 4x + 3$



- The line slants sharply down from left to right.
- The line passes through the following points:
 - (0, 8)
 - $(\frac{11}{2}, 0)$

The graph of the linear function f is shown, where $y = f(x)$. What is the y -intercept of the graph of f ?

- A. 0,0
- B. 0, $-\frac{16}{11}$
- C. 0, -8
- D. 0,8



- The line slants sharply up from left to right.
- The line passes through the following points:
 - (0 comma negative 4)
 - (1 comma 0)

The graph of the function f is shown, where $y = f(x)$. What is the y-intercept of the graph?

- A. 0, -1
- B. 0, -4
- C. 0, 1
- D. 0, 4